

The U.S. Housing Problem as a System: From Dirt to Keys, With Pittsburgh as the Laboratory

The most important conclusion from the research is that **“America needs more housing” is true, but it is too coarse a problem statement to design against.**

There are at least five different shortages interacting at once: a shortage of units in high-demand places, a shortage of units affordable to particular incomes, a shortage of *buildable/entitled* land even where raw land exists, a shortage of infrastructure or capacity to serve projects, and—since rates rose sharply—a shortage of *existing homes actually coming onto the market*. Those problems look similar to a buyer facing a high price, but they have very different causes and therefore very different technical interventions. Freddie Mac estimated the national structural housing shortage at about 3.7 million units as of Q3 2024, while Pennsylvania now projects that, absent further action, it will fall roughly 185,000 homes short of need by 2035. ¹

The 2026 market demonstrates why the distinction matters. In August, U.S. housing starts ran at a 1.275 million annualized pace and completions at 1.128 million, with completions 27.1% below a year earlier. Yet the *new-home* market simultaneously had 483,000 homes for sale, equivalent to 8.5 months of supply at the current sales rate. New single-family sales were running at 684,000 annually. In other words, you can have a long-run housing shortage, severe affordability problems, weak turnover of existing homes, **and plenty of newly built inventory at a particular price point at the same time.** ²

That distinction should shape your hackathon approach. The three sponsor briefs are actually pointing at three different layers of the same operating system:

- **Challenge 01** is about *project uncertainty*: “Can I build this here, and what will stop me?” It asks for parcel-level zoning, infrastructure, environmental and policy reasoning, with explicit human-review points. `filecite turn0file0`
- **Challenge 02** is about *system observability*: “What is actually being proposed, permitted, completed, rented and lost?” It explicitly identifies fragmented and lagged data as the problem. `filecite turn0file1`
- **Challenge 03** is about *decision quality*: “Even if I can build, what should I build here?” It asks teams to distinguish evidence from normative choices and compare typologies rather than proclaim a single answer. `filecite turn0file2`

My strongest thesis after going through the evidence is this:

A surprisingly large housing opportunity is not “AI designs houses.” It is reducing the cost of uncertainty between deciding to build a house and knowing that you are actually allowed, able, financed and ready to build it.

That uncertainty is expensive because land and capital sit idle while lawyers, architects, engineers, planners, utility reviewers, lenders, boards, inspectors and developers exchange information—often serially, often through differently structured systems, and often while nobody has a complete dependency map.

The real problem is several housing markets, not one

Start with a better mental model of “supply”

Imagine five nested inventories:

Physical housing stock → usable housing → legally usable housing → financially attainable housing → actually available housing.

A city can have plenty of physical structures while still having a severe shortage at the final two layers.

Pennsylvania illustrates this particularly well. More than half of the Commonwealth's housing stock is over 50 years old, more than one million Pennsylvania households spend over 30% of income on housing, and the state says about 450,000 additional units will be needed by 2035 while current construction trajectories imply an approximately 185,000-unit gap. ³

Pittsburgh is not Phoenix, Austin or Miami. Its challenge is partly **old stock, disinvestment, difficult terrain, fragmented parcels, legacy infrastructure and affordability mismatch**, rather than simply rapid population growth outrunning greenfield subdivision production. The hackathon's own first brief specifically describes Pittsburgh's “older housing stock and constrained development environment” and focuses on starter and affordable housing feasibility. [filecite turn0file0](#)

That matters because “build more” branches into several different questions:

Question	What may actually be scarce?	Possible intervention
Why can't this household buy?	Monthly payment capacity	Rates, financing, down-payment support, price
Why isn't a developer building?	Feasible residual economics	Entitlement, land, infrastructure, financing, construction cost
Why is an empty lot still empty?	Legal/buildable capacity	Zoning, geometry, title, utilities, environmental conditions
Why is an existing owner not selling?	Mobility / replacement-home economics	Mortgage lock-in
Why aren't affordable units getting built?	Subsidy / operating gap	LIHTC, vouchers, PHARE, local gap financing
Why is permitting slow?	Information processing / queueing / rework	Better intake, rules engines, parallel workflows

Question	What may actually be scarce?	Possible intervention
Why does nobody know what's happening?	Data integration	Project/parcel event graph
What type should go here?	Matching demand to physical/legal constraints	Scenario analysis

This is also why the sponsor's "fragmented data" statement is more profound than it initially sounds.

Housing is fundamentally a joins problem. Parcel geometry lives in one system. Ownership and assessments live elsewhere. Zoning is another layer. Permits are applications linked imperfectly to addresses and parcels. Environmental records use different geographies. Utilities use networks rather than parcels. Household need is measured at census geography. Rents may come from private listings. Financing happens behind closed doors.

Challenge 02 recognizes almost exactly this: it asks for movement from region → tract → parcel → building → project while making missingness, reporting bias and privacy visible rather than pretending the joins are perfect. [filecite turn0file1](#)

There is another hidden supply: "option value"

A developer does not need merely a parcel. They need a parcel with a credible option to become a project.

Suppose two pieces of land each cost \$500,000:

- Parcel A is zoned appropriately, has adequate water/sewer capacity, no flood issue, straightforward access and predictable permits.
- Parcel B *might* support the same building but requires a variance, a sewer study, unclear stormwater mitigation, utility coordination and perhaps a board hearing.

The physical land supply is two parcels. The **financeable buildable supply may effectively be one**.

This is why your sponsor's proposed Development Ease Score is potentially useful—but only if it is not a black-box score. The brief itself stresses source grounding, explanations, assumptions and human-review points. [filecite turn0file0](#)

A much more valuable output than:

Development Ease Score: **72/100**

would be:

Likely by-right: medium confidence
Earliest plausible permit-ready date: 122–185 days
Critical-path uncertainty: sewer capacity
Next irreversible spend: \$X engineering study
Potential zoning variance: none found

External dependencies: Pittsburgh Water, PA DEP

Evidence freshness: zoning 3 days old; utility capacity unknown

Estimated cost of a 30-day delay: \$Y under user's financing assumptions

Human confirmation required: utility capacity, title easements

That is no longer an "AI score." It is a primitive **development operating system**.

The developer journey from parcel to keys

A developer building at scale is really operating a sequence of expensive options. At each stage they spend more money to reduce uncertainty about whether it is rational to spend the next dollar.

The process is not strictly linear, but conceptually it looks like this:

Market thesis → site → title/due diligence → concept → entitlement → financing → permit documents → permits → construction → inspections → occupancy → lease/sale

The dangerous part is that many of those arrows can loop backward.

Site selection and market thesis

Before anyone draws a building, the developer asks:

What could be sold or rented here, to whom, at what achievable price, and at what velocity?

For a market-rate project, this involves sales or rent comparables, household growth, incomes, vacancies, concessions, competing pipeline, absorption, financing assumptions and exit values.

For an affordable project, the equation adds income restrictions, allowable rent, vouchers, operating subsidies, tax credits, grants, soft loans and affordability periods. Pittsburgh's local Housing Opportunity Fund currently supports mechanisms including Rental Gap, For-Sale Development, down-payment and closing-cost support and other housing programs; the City committed \$10 million annually for 12 years beginning in 2018. ⁴

The fact that "gap financing" exists is itself an important clue: **a project can be physically useful, socially needed and legally permissible while still not pencil out financially.**

For example, in May 2026 the URA considered Rental Gap Program support for three Pittsburgh projects totaling 159 units, 149 of them affordable. ⁵ Pennsylvania's mixed-use housing pilot is another striking signal: 104 applications sought more than \$469 million from a program with only \$10 million available. That does not prove every request represented an otherwise viable project, but it shows an enormous demand for capital intended to close development feasibility gaps. ⁶

Site control and title

Once a site looks interesting, you want to know **what you are actually buying**.

Questions include:

Who owns every parcel? Are there liens? Easements? Restrictions? Rights-of-way? Encroachments? Access? Separate parcels that need consolidation? Are the legal description and physical boundaries consistent? Is some “vacant” strip actually owned by another party?

For sophisticated transactions, the 2026 ALTA/NSPS Land Title Survey standards provide a standardized way to align surveyors, lenders, title insurers and clients around boundaries, easements, encroachments, access and related title matters. The new standards became effective February 23, 2026. ⁷

In Pittsburgh, a subdivision or consolidation plan must be prepared by a licensed surveyor; major subdivisions and consolidations go to the Planning Commission, and approved plans must subsequently be recorded with Allegheny County. ⁸

Notice the intelligence chain already:

developer → attorney/title → surveyor → city planning → county recording → back to city records.

Each actor is rational. The system as a whole can still be slow.

Physical due diligence: what is really under the ground?

This is where a pretty GIS map meets reality.

A scalable development may need some combination of:

Investigation	What it protects against
Boundary / ALTA survey	Boundary, access, easement and encroachment errors
Topographic survey	Elevation, slope, grading and drainage surprises
Utility survey / records	Unknown lines, easements and connection points
Geotechnical investigation	Soil bearing, fill, rock, mine conditions, retaining requirements
Phase I Environmental Site Assessment	Potential contamination/liability
Phase II testing	Actual sampling where Phase I identifies concerns
Flood analysis	Floodplain requirements and insurability
Wetlands/waters review	Federal/state environmental constraints
Stormwater study	Detention/infiltration/impervious-area requirements
Sewer/water capacity	Whether the project can actually be served

Investigation	What it protects against
Traffic/transportation study	Network/access impact and mitigation
Existing-building investigation	Structure, hazardous materials, MEP condition for reuse

A Phase I ESA is not simply “some environmental report.” EPA recognizes ASTM E1527-21 as consistent with federal All Appropriate Inquiries requirements, which can matter for certain CERCLA liability protections.

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At a large enough construction disturbance, Pennsylvania introduces another layer. Construction earth disturbance of an acre or more can trigger Chapter 102 NPDES permitting. PA DEP has itself experimented with expedited processes; one of its permit modernization pilots said it expected to cut average review time for certain Chapter 102 individual NPDES permits by 73 business days.

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That is a crucial data point for your thesis: **government is explicitly acknowledging that elapsed review time is a material economic variable.**

Zoning and development rights

Next comes the deceptively simple question:

What can I legally put here?

Not merely “what color is the zoning map?”

You need to evaluate use, units, lot size, density, setbacks, height, lot coverage, parking/loading, access, overlays, floodplain, historic status, design requirements, affordable housing provisions, subdivision, and potentially discretionary approvals.

Pittsburgh's zoning workflow has multiple review levels—Basic Zoning Review, Site Plan Review and review involving the Planning Commission—and additional notices, hearings or reviews can arise depending on location and scope.

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This is where regulation can create absurd results through interacting rules rather than one obviously bad rule. Pittsburgh recently eliminated minimum lot size **per dwelling unit** and reduced other minimum lot sizes. Before that reform, for example, R2-M zoning could permit a duplex as a use but still require a variance on a 3,200-square-foot parcel because of the per-unit lot-area rule. The city's reform removed that particular collision.

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That is a perfect “AI opportunity” pattern:

Rule A says yes.

Rule B says yes.

Rule C, applied to the geometry, quietly makes the project impossible.

Humans are good at eventually discovering these contradictions. Software is potentially much better at discovering them **before someone buys the land**.

Pre-application and discretionary review

For sufficiently complex Pittsburgh projects, a developer may go through a pre-application meeting with City Planning. It is recommended for complex Site Plan or Planning Commission projects and can be required at thresholds such as new commercial structures/additions of at least 5,000 square feet, more than 20 parking spaces, master plans/rezonings and certain other situations. The current fee is \$250. ¹³

This is not necessarily bureaucratic waste.

A pre-application meeting is a form of **human uncertainty compression**. Experienced planners can detect “you don’t realize this requirement will apply” before an applicant spends months on a bad design.

That creates an important product insight:

Don’t automatically ask, “How do we eliminate the planner?”

Ask, “How do we encode the 80% of predictable planner knowledge so the planner spends time on the 20% genuinely requiring judgment?”

Utility capacity and infrastructure

A zoning-compliant project is not necessarily serviceable.

Pittsburgh Water asks developers to engage early. Development workflows can involve availability determinations, pre-development coordination, professional-engineer-stamped tap-in plans and, where applicable, Pennsylvania sewage-facilities planning. The utility explicitly advises starting early because its approvals can interact with building permitting. ¹⁴

This is another major opportunity for a “Development Ease” product. A parcel score based on zoning but not infrastructure is dangerously incomplete.

You want the application to answer:

Can I build?

then

Can I connect?

then

At what incremental infrastructure cost?

then

Who must approve that answer?

Those are different questions.

Stormwater and land disturbance

Pittsburgh requires a commercial stormwater permit for projects involving at least 10,000 square feet of land disturbance, at least 5,000 square feet of increased impervious surface, and certain lower-threshold riverfront cases. ¹⁵

Separate land-operations requirements can arise for major grading, work on steep slopes, large paved areas, drainage-channel changes and related activity. ¹⁶

A useful software system therefore needs **trigger logic**, not just static data:

```
IF land_disturbance >= 10,000 sq ft
THEN stormwater review likely triggered

IF earth_disturbance >= 1 acre
THEN check PA Chapter 102 / NPDES pathway

IF subdivision needed
THEN survey → city approval → county recordation

IF utility demand changes
THEN utility development workflow

IF discretionary zoning threshold triggered
THEN hearing/review path changes
```

That “requirements compiler” may be substantially more useful than generating architectural renderings.

The permit package

Pittsburgh is actively changing this system right now.

The new Building and Development Application combines what had been separate zoning-development and building-permit applications. Relevant reviews can flow to PLI, City Planning and DOMI, and if a DOMI permit is needed, a reviewer can create the sub-permit. ¹⁷

That is important because a hackathon team using an outdated model of Pittsburgh permitting could accidentally build a solution to a problem the City has already begun solving.

The official current permit process is broadly:

Application submission → application review → permit issuance → inspections → completion.

OneStopPGH determines required documents based on permit/structure/work configuration and handles revised submissions. ¹⁸

But review time is still consequential. Pittsburgh's standard permit reviews can carry service-level targets in the 15–30-business-day range, and the City's 2026 EZ Permit pilot makes the contrast explicit: eligible low-risk scopes can move from plans plus 15–30 business days of review to **no plan review and approximately one business day**, with applicants accepting responsibility through attestation. ¹⁹

This is one of your strongest empirical clues.

The City has effectively classified some reviews as:

“Human review adds less expected safety value than it costs in delay for this low-risk class.”

That suggests a general research question:

Which other housing decisions are being given expensive bespoke human attention despite being sufficiently standardized to use rules + attestation + audit?

Construction is not the end of regulation

After permits, there are construction-phase inspections.

Some work also requires special inspections. Pittsburgh states explicitly that special inspections **supplement rather than replace** PLI inspections. The design professional identifies applicable special inspections, the owner hires qualified special inspectors, and the permit holder must coordinate both inspection layers and may have to leave work exposed until both have cleared it. ²⁰

Again, don't immediately call that duplication “waste.” Structural inspection exists because failure can kill people.

But it is a great place to ask:

- Are both inspectors collecting overlapping evidence?
- Can one photo/test result satisfy multiple documentation needs?
- Can inspection prerequisites be scheduled automatically?
- How often does work remain exposed waiting for a second inspection?
- How much reinspection occurs because the wrong inspection was requested?
- How many days of schedule variance come from coordination rather than actual defects?

That is design thinking.

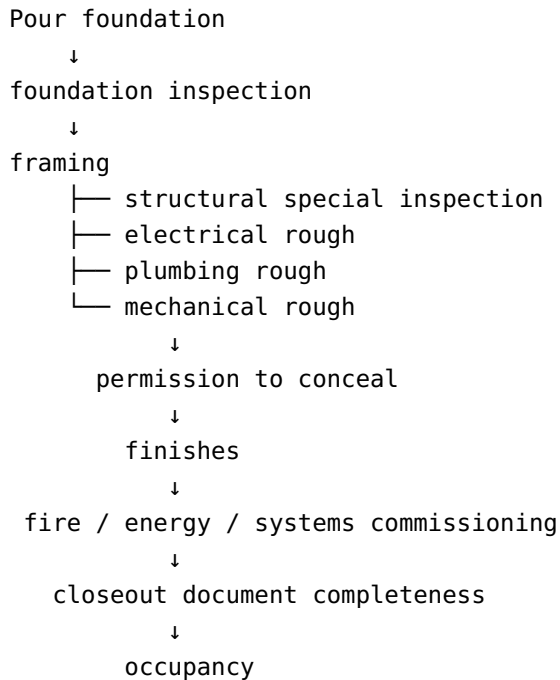
Close-out and occupancy

At the end, projects can still be delayed by testing reports, fire-system documentation, energy/commissioning records, special-inspection reports and administrative closeout before final occupancy. Pittsburgh's process includes administrative review of required closeout documents before issuance of the final certificate. ²¹

A developer therefore doesn't merely need a checklist of permits.

They need a **dependency graph with evidence requirements**.

Conceptually:



Every unresolved edge can become a lost day.

The buyer journey from search to closing

The buyer's side is a different maze. The developer battles feasibility; the homebuyer battles **information asymmetry, financing and transaction coordination**.

A simplified journey is:

Budget → mortgage shopping → search → offer → inspection → appraisal → title/legal checks → underwriting → final disclosure → closing → taxes/insurance/ownership

Start with the payment, not the house price

Buyers frequently shop in terms of a price—"I can buy a \$400,000 house"—but their lender ultimately cares about the financial package.

At the moment, this distinction is enormous. Freddie Mac's national average 30-year fixed mortgage rate was **7.03% on September 24, 2026**, compared with 6.30% one year earlier. ²²

Take a \$400,000 property with 20% down, so the loan is \$320,000.

At 3% for 30 years, principal and interest are approximately:

\$1,349/month

At 7%:

\$2,129/month

The same principal therefore costs about **\$780 more every month**, roughly 58% more, before taxes, homeowner insurance, HOA charges or maintenance.

This is why high rates can create an affordability crisis even without rapidly rising home prices.

Mortgage shopping and underwriting

A lender must generally provide a Loan Estimate within three business days once it has the six core application items required under the mortgage disclosure framework. CFPB encourages consumers to compare multiple Loan Estimates. ²³

Once the buyer proceeds, underwriting becomes document-heavy:

income, assets, debts, credit, employment, down payment, source of funds, property valuation, insurance, title conditions and potentially more.

The buyer and lender are simultaneously underwriting two different things:

Can this human repay?

and

Is this property adequate collateral?

That separation explains why an appraisal and an inspection are not substitutes.

The **inspection** is primarily for the buyer—condition, defects, systems, repairs.

The **appraisal** is primarily a valuation/collateral function for the lender. CFPB explicitly warns buyers that a lender-required appraisal does not replace a home inspection. ²⁴

Title and property legality

Now you encounter many of the same data systems as the developer—but from the other direction.

A buyer needs confidence that:

- the seller can convey title;

- liens/easements/issues are known;
- the physical thing being purchased corresponds to the legal parcel;
- existing use is legitimate;
- insurance can be obtained;
- taxes and assessments make sense.

Pittsburgh has a particularly revealing local requirement: a **Property Certification** is required for a real-estate closing due to sale or refinance. The certificate reports zoning classification, legality of use, historic status, occupancy information and active code violations. It currently costs \$100. ²⁵

But a subtle historical-data problem appears immediately.

Pittsburgh does **not** require a Certificate of Occupancy merely to buy or sell a property, and older single-family houses may have no CO because historic regulations did not require one. The absence of a record therefore does not mean either “illegal” or “compliant.” ²⁶

This is precisely the kind of situation an AI system routinely mishandles.

Bad AI:

“No occupancy certificate found → property noncompliant.”

Good system:

“No occupancy certificate found. Pittsburgh notes that older single-family properties may legitimately lack one. Current legal-use status requires additional verification. Confidence: low.”

That difference is the difference between search and reasoning.

Inspections and contingencies

The buyer may inspect the home and use contract contingencies to renegotiate, demand repairs or exit depending on the agreement and findings. Lenders separately evaluate the collateral. ²⁴

Now imagine Pittsburgh's older housing stock: roof, foundation, knob-and-tube wiring, sewer lateral, retaining walls, lead, water infiltration, mechanical systems, unpermitted additions, hillside conditions and energy performance can all change the economics.

This suggests a completely different AI product opportunity:

A “true cost of ownership” layer that converts inspection findings and public records into a five- or ten-year capital-needs forecast rather than merely summarizing an inspection PDF.

The caution is obvious: it should expose assumptions and ranges rather than pretend to diagnose a house remotely.

Closing costs contain another maze

Nationally, typical mortgage closing costs are often described as roughly 2–5% of the purchase price, excluding the down payment, though the actual number varies greatly by transaction. Title services can be a major third-party component. ²⁷

Pennsylvania imposes a 1% state realty transfer tax. ²⁸

Here is a fascinating Pittsburgh-specific data quality problem I found while researching your idea:

Allegheny County, which collects the tax through the recording process, says Pittsburgh transactions are subject to:

- 1% Commonwealth
- 3% City of Pittsburgh
- 1% Pittsburgh School District
- **5% total**

and specifically says the overall 5% Pittsburgh rate has applied since 2020. ²⁹

Yet the **City of Pittsburgh's own tax webpage**, marked updated March 5, 2026, currently lists:

- 1% Commonwealth
- 2% City
- 1% School District
- **4% total**

³⁰

That is not a theoretical “fragmented data” complaint. **Two official government pages presently disagree about a financially material transaction rate.** The county Recorder's operational page indicates 5% and its local-rate table lists Pittsburgh at 3% municipal plus 1% school, but a buyer or software system should not silently decide which page is “truth”; the contradiction should be surfaced and the operative rate verified in the transaction. ³¹

This is perhaps the most hackathon-relevant little discovery in the entire research exercise.

AI's role should be:

“I found two authoritative sources that conflict. Here is the conflict, timestamps, likely operational source, and person/agency responsible for final confirmation.”

Not:

“The answer is definitely X.”

Final disclosures and closing

For most covered mortgages, the lender must provide the Closing Disclosure at least three business days before closing. CFPB recommends comparing it with the latest Loan Estimate and resolving unexpected changes before signing. ³²

Even after purchase, another Pittsburgh-specific uncertainty exists: property assessments can be appealed by owners **or taxing bodies**, and Allegheny County has a formal administrative and court appeal chain. ³³

So the buyer's final “price” is not merely:

purchase price + mortgage.

It is better thought of as:

acquisition cash + financing cost + taxes + insurance + immediate repairs + expected capital expenditures + transaction costs + risk premium for unknown conditions.

There is an enormous information-product opportunity in making that legible.

Where time and money actually disappear

The obvious place to look is lumber, concrete and labor.

Those matter, but a building has another category of cost that behaves almost like invisible material:

time exposed to capital.

Hard costs are only one piece

NAHB's 2022 survey of single-family builders estimated that construction represented about 60.8% of final sale price and the finished lot another 17.8%, with the remainder including financing, overhead, marketing, sales commissions and profit. Within construction itself, interior finishes, framing and major systems were large components. Treat those numbers as industry survey evidence rather than universal engineering constants. ³⁴

NAHB's 2026 regulatory-cost study—a builder-industry estimate that should be treated as such—put costs attributable to regulation at approximately \$131,734, or 26.4% of its roughly \$499,500 average new single-family home price, including both land-development and construction-stage effects. ³⁵

That number is politically contested territory because defining the “cost of regulation” requires judgments about what counterfactual development would look like. You should **not** bake “26.4%” into a product as ground truth.

Instead, instrument individual projects.

Measure:

- What fee was paid?
- How many consultant hours were required?
- What redesign happened?
- How many calendar days were consumed?
- What additional construction was required?
- What safety/environmental/public value did the requirement create?

Then let the data tell you the local number.

Cost-of-delay is potentially bigger than the visible permit fee

Suppose a developer has \$10 million of capital exposed to a project at an illustrative 8% annual carrying cost.

That capital costs roughly:

\$2,192 per calendar day

or

\$66,667 per month

A 60-day avoidable delay therefore represents roughly:

\$131,500

in financing carry alone—before staff overhead, option extensions, contractor escalation, redesign or lost rent.

That is why optimizing a \$1,000 fee while ignoring three months of uncertainty is often economically backwards.

A very powerful hackathon metric would therefore be:

Expected Cost of Delay = probability of delay × expected delay days × daily capital exposure

Different developers can insert their actual financing rate and capital exposure.

Now the “Development Ease Score” becomes dollars.

The housing “idiot index”

You mentioned the **Idiot Index**. The term associated with Elon Musk is generally described as:

finished component cost ÷ raw-material cost.

It is intended as a first-principles prompt: if raw aluminum is worth \$10 but the part is \$1,000, what exactly created the other \$990? The concept has been popularized from accounts of SpaceX/Tesla manufacturing; it is not a recognized housing-economics metric. ³⁶

Applying it literally to a house would be misleading.

A house is not merely timber + steel + concrete. A large portion of legitimate value lies in:

labor, land, infrastructure, engineering, architecture, equipment, logistics, safety verification, financing, legal certainty and location.

But the **spirit** of the Idiot Index is extremely useful.

I would create a *Housing Friction Index* instead:

$$\text{Friction Wedge} = \frac{\text{avoidable coordination} + \text{delay} + \text{rework cost}}{\text{all-in development cost}}$$

And then several operational measures beneath it:

Metric	Question it answers
Queue-to-touch ratio	How many days does a submission wait per hour of actual review?
First-pass yield	What share of applications clear a review without resubmission?
Rework ratio	How much professional work is spent responding to preventable completeness issues?
Data re-entry count	How many times is the same parcel/project fact manually entered?
Decision density	How many human approvals exist per \$1M or per housing unit?
Seriality factor	How many approvals that could run in parallel instead run sequentially?
Unknown-dependency count	How many critical requirements become visible only after spending money?
Cost-of-delay	How much financing/overhead cost is generated by elapsed process time?
Evidence reuse rate	How often can one verified document satisfy multiple agencies?
Rule-exception rate	How frequently does the formal code fail to predict the actual path?

That is essentially the **idiot index for bureaucracy**, but it does not start from the insulting assumption that the bureaucracy is idiotic.

It starts from measurement.

Separate necessary intelligence from coordination waste

This distinction is critical.

Human layer	Why it exists	What AI can realistically attack
Surveyor	Establish legal/physical boundaries	Records extraction, conflict detection, document prep—not field/legal certification
Geotechnical engineer	Ground conditions can destroy structures	Prior-risk screening, data aggregation—not replacing boring/testing
Planner	Interpret land-use rules/context	Rule compilation, precedent search, completeness
Architect/engineer	Life safety and integrated design	Code checking, clash detection, documentation QA
Permit reviewer	Independent compliance verification	Triage, low-risk automated pathways, inconsistency detection
Public board	Exercises discretionary/public judgment	Better evidence and scenario visualization, not “AI votes”
Utility engineer	Protect scarce network capacity	Automated capacity pre-screening
Inspector	Verify the thing built matches approved/safe condition	Scheduling, evidence capture, computer-vision assist
Lender/underwriter	Allocates financial risk	Data reconciliation and scenario analysis
Appraiser	Independent collateral valuation	Comparable search and anomaly detection
Title professional	Resolves legal ownership risk	Search and document extraction
Buyer/home inspector	Protect purchaser against hidden condition	Structured findings and repair-cost scenarios

The target is not “remove humans.”

The target is:

stop expensive humans from spending their time finding documents, copying addresses, checking obvious thresholds, chasing missing forms, explaining status, or rediscovering dependencies.

Pittsburgh's EZ Permit experiment supports exactly this logic: low-risk scopes move to applicant attestation and roughly one-day issuance, while higher-risk work retains professional review. ¹⁹

That is a much more defensible AI narrative than “replace city planners with an LLM.”

Pittsburgh is an unusually good living laboratory

Pittsburgh is useful because it contains almost every category of housing friction in a manageable geographic area.

It has older stock, steep terrain, rivers/floodplain, legacy industrial sites, aging infrastructure, fragmented ownership, major institutional landowners, neighborhood political structures, affordable-housing gaps, relatively inexpensive properties in some neighborhoods, high development costs, adaptive-reuse opportunities and a permitting system undergoing reform.

Pennsylvania's first statewide Housing Action Plan now explicitly lists modernizing development regulation and improving interagency coordination/data among its five central goals. The plan was based on more than a year of engagement, including 18 regional roundtables and nearly 2,500 survey responses from all 67 counties. ³⁷

Meanwhile Pittsburgh's own government issued an executive order in January 2026 declaring its existing permitting system “outdated” and directing departments to simplify, streamline and make approvals more predictable. ³⁸

And then the City started shipping changes: a new permit tracker, Building and Development Application consolidation, EZ Permits and work to modernize transportation-impact review. ³⁹

This means you have an unusually good opportunity to study **before and after**.

Pittsburgh's particularly interesting friction points

Transportation studies are one example. DOMI itself said in 2026 that Transportation Impact Studies were being submitted separately from the rest of the development application, which could create longer review times, evolving requirements and uncertainty; the department is now redesigning that process. ⁴⁰

Special inspections provide another example: multiple independent verification layers can be necessary, but poor coordination can keep work uncovered and crews waiting. ²⁰

The City's property-sale process is another. Pittsburgh created a Home Buying Pilot after recognizing that the legal and legislative process associated with purchasing certain city-owned properties was too long and complicated for typical lower- and moderate-income homebuyers. ⁴¹

Infrastructure is another. Pittsburgh Water's development process includes pre-development coordination and engineering documentation, and the developer manual explicitly contemplates ongoing dialogue with development staff rather than treating the rules as completely self-executing. ⁴²

And old records are another. The City openly acknowledges that some older single-family properties simply will not have modern occupancy records. ²⁶

These are not four separate software problems.

They are manifestations of:

The state of a housing project cannot be inferred from any single database.

Who I would interview before the challenge

Do not begin by asking people, “Would an AI permit tool help?”

That question invites politeness and solution bias.

Instead, hand them a real project and say:

“Walk us from the moment you first heard about this property until somebody got keys. Every time you waited, changed a drawing, paid someone, called someone, learned something unexpectedly or submitted the same information twice, stop us.”

I would prioritize these Pittsburgh groups:

Interview	Why it matters	What to ask them to show you
City of Pittsburgh PLI / permit-reform team	Sees permit queues and failure patterns	Five applications that went smoothly and five that bounced
Department of City Planning / Zoning	Knows where code becomes judgment	Cases where map/code lookup was insufficient
DOMI development-review team	External dependency / transportation	A TIS/project that went through multiple revisions
Pittsburgh Water development services	Infrastructure feasibility often discovered late	Actual development dependency chain from availability through taps
Allegheny County Health plumbing reviewers	Cross-jurisdiction layer	Where City/County sequencing creates applicant confusion
URA Housing / Housing Opportunity Fund	Sees projects that are needed but financially infeasible	Real pro formas before/after gap financing
ACTION-Housing	Pittsburgh's largest nonprofit developer and actual repeat user	One affordable project from acquisition through occupancy

Interview	Why it matters	What to ask them to show you
Builders Association of Metropolitan Pittsburgh	Can recruit multiple private builders quickly	Ask 5 builders to rank recurring delay/cost causes
Pittsburgh Community Reinvestment Group	Community/housing-policy counterweight to developer view	Where “streamlining” risks displacement or public harm
City of Bridges Community Land Trust	Permanent affordability / homeowner perspective	Where acquisition and resale mechanics create hidden friction
Local surveyor + civil engineer + geotech	They find the “unknown unknowns”	What developers repeatedly fail to discover early
Title attorney / title company	Sees ownership/legal-data failures	Three deals delayed by title rather than construction
Mortgage loan officer + buyer agent + home inspector	Buyer-side reality	What buyers learn too late
Small developer, not just major developer	Most likely to be crushed by fixed process cost	How much expertise is effectively required to submit correctly

ACTION-Housing is especially useful because it describes itself as Pittsburgh's largest nonprofit developer and works in both housing development and housing services. ⁴³ BAMP represents nearly 400 member businesses, giving you a potential channel to production builders, remodelers, suppliers and trades rather than relying on one anecdote. ⁴⁴ PCRG says it works with 65 community organizations covering 125 neighborhoods, giving you the opposite but equally necessary community perspective. ⁴⁵

Your interview question should always be “show me”

For the developer:

Show me the last project where you discovered something too late.

For the planner:

Show me an application where a competent developer misunderstood the code.

For the permit reviewer:

Show me the most common reasons applications return for revision.

For the civil engineer:

Show me something invisible on the zoning map that changed feasibility.

For Pittsburgh Water:

Show me the earliest point at which you can confidently say this development is serviceable.

For the affordable-housing developer:

Show me the gap between achievable rent and the rent required for the project to finance.

For the buyer:

Show me every number you did not know when you made the offer.

For the inspector:

Show me the defects that public data could have predicted—and the ones it never could.

For the lender:

Show me exactly when a “good” project stops penciling because rates or construction costs changed.

For every person:

What fact, had you known it 90 days earlier, would have changed your decision?

That last question may give you the entire product.

Interest rates, money and the housing machine

Your instinct from Ray Dalio's *Changing World Order* framework is useful, but there is a danger of jumping too quickly from:

“fiat money / debt cycles”

to

“therefore housing is expensive.”

The actual mechanism is much more interesting.

The housing monetary-transmission chain

A simplified chain is:

Federal Reserve / inflation expectations / Treasury market

↓

long-term yields + mortgage-backed-security pricing

↓

mortgage rates and construction/development financing

↓

buyer monthly payment + developer required return

↓

home demand + land values + project feasibility

↓

starts / sales / prices

But then U.S. fixed-rate mortgages create a feedback loop:

higher current rates

↓

existing owner says "I have a 3% mortgage; why would I sell and take a 7% mortgage?"

↓

fewer existing homes listed

↓

less transaction supply

↓

prices can remain unexpectedly firm even while demand weakens

That last part has become extremely important.

FHFA research estimates that for every one-percentage-point increase in the gap between prevailing mortgage rates and a homeowner's existing mortgage rate, the probability of sale falls about **18.1%**. Its later geographic analysis estimated mortgage lock-in prevented roughly 1.72 million sales from 2022 Q2 through 2024 Q2 and estimated that the resulting supply reduction added about 7% to house prices, offsetting a roughly 5.6% direct price-reducing effect from higher rates. These are model-based estimates, not universal mechanical laws, but they explain the current paradox very well. ⁴⁶

Federal Reserve researchers independently estimate that mortgage lock-in explained about **44% of the decline in mortgage-borrower mobility from 2021 to 2022**, primarily by reducing local moves. ⁴⁷

And the Fed's July 2026 Monetary Policy Report noted that a majority of outstanding mortgages still carried rates below 4%, while prevailing new mortgage rates remained substantially higher. ⁴⁸

So rising rates hit the housing market in several directions simultaneously:

Effect of higher rates	Direction
Buyer monthly payment	↑
Buyer purchasing power	↓

Effect of higher rates	Direction
Demand	↓
Price pressure from demand	↓
Builder/developer financing cost	↑
Construction feasibility	↓
New supply response	↓
Existing-owner willingness to sell	↓
Existing-home inventory	↓
Price pressure from reduced resale supply	↑
Rent demand as would-be buyers remain renters	potentially ↑

That is why saying “rates up → home prices down” is an inadequate model.

Monetary policy really does reach housing

This is not merely correlation.

Federal Reserve research finds monetary policy significantly affects homeownership choices and housing rents. ⁴⁹

Research on Federal Reserve large-scale asset purchases found that purchases of agency mortgage-backed securities and Treasuries lowered MBS yields and mortgage rates beyond what changes in market expectations alone would imply. ⁵⁰

A BIS study using high-frequency monetary-policy surprises found that a surprise that raised 30-year mortgage rates by 25 basis points reduced listing prices by roughly 1% within two weeks in its sample, with the effect carrying through substantially to eventual transaction prices. ⁵¹

So there is absolutely a monetary layer to the housing puzzle.

But “fiat money caused the housing shortage” goes too far

Housing combines **financial scarcity** and **physical scarcity**.

The central bank can affect:

- cost of credit;
- asset discount rates;
- mortgage demand;
- balance sheets;
- liquidity;

- risk appetite.

It cannot directly create:

- a serviced lot;
- a sewer main;
- a carpenter;
- a transformer;
- a zoning entitlement;
- a completed structural inspection;
- a parcel with clean title.

This creates an inference that I think is the useful Dalio connection:

When financial demand expands rapidly while the physical/legal supply system is inelastic, more purchasing power can capitalize into land and existing-home prices rather than rapidly producing more units. Conversely, when rates rise sharply, demand falls—but construction and resale supply can also contract, preventing the clean price correction a simple demand model predicts. ⁵²

San Francisco Fed research adds an important correction to the popular narrative: it found that the extraordinarily low inventory of 2020–22 was driven substantially by unusually strong demand and rapid sales rather than by a collapse in new listings; rising rates then cooled demand and helped inventory recover. ⁵³

So there are really **three clocks**:

1. **Money clock** — rates can move within days.
2. **Demand clock** — buyers react within weeks/months.
3. **Housing-supply clock** — entitlement, infrastructure and construction can take years.

That timing mismatch is fundamental.

This creates another technology opportunity: dynamic feasibility

Most conventional pro formas are snapshots.

But a project may take years from acquisition to stabilization. You actually want:

Feasibility = $f(\text{rent/sale price, construction cost, interest rate, entitlement probability, schedule, infrastructure cost})$

And each one is a distribution, not a fixed number.

An AI/analytics system could say:

At today's assumptions this project produces a 6.1% yield on cost.
If entitlement slips 120 days and debt costs rise 50 bps, it falls to 5.4%.
The highest-value uncertainty to resolve next is sewer capacity.

Spending \$8,000 to resolve that uncertainty has higher expected value than completing architectural design.

That is genuinely intelligent decision support.

It tells the developer **what to learn next**.

What I would build and what I would validate before the hackathon

I would not build another housing dashboard first.

I would also not build “ChatGPT for zoning.”

I would build a **Parcel-to-Keys Evidence Graph** that happens to produce the sponsor's Development Ease Score.

The product concept

Input:

Parcel ID + proposed housing concept.

For example:

“32 townhomes”

or

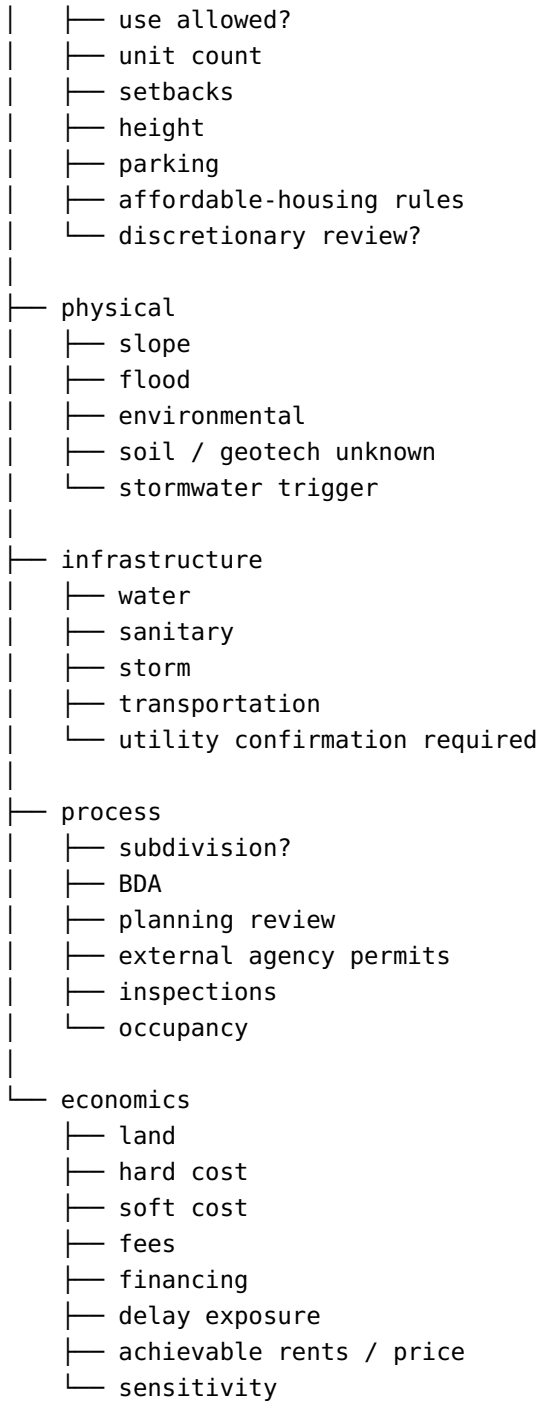
“60-unit apartment building, 20% affordable”

or

“duplex on existing residential lot.”

The system creates a project-specific graph:

```
PARCEL
|
├─ title / ownership
|   ├── owner
|   ├── legal description
|   ├── easements
|   └── consolidation required?
|
└─ zoning
```



Every node contains:

**status → evidence → source → timestamp → confidence → owner → prerequisite → human reviewer
→ estimated duration → estimated cost.**

That architecture unifies all three sponsor tracks.

Challenge 01 contributes the **feasibility / process graph**. [filecite turn0file0](#)

Challenge 02 contributes the **historical event/data layer** from previous projects. [filecite turn0file1](#)

Challenge 03 contributes **alternative housing scenarios and explicit weighting/tradeoffs**.
[filecite turn0file2](#)

The Development Ease Score should be the least interesting output

Something like:

$$DES = f(Z, I, E, T, R, D, F)$$

where:

- Z = zoning uncertainty
- I = infrastructure uncertainty
- E = environmental/physical uncertainty
- T = title/site-control uncertainty
- R = review complexity
- D = expected delay
- F = financing sensitivity

But never hide them behind one number.

Show:

68/100 overall

Zoning: 92

Infrastructure: 41

Environmental: 78

Title: 96

Process predictability: 63

Financial resilience: 54

Primary reason the score is low: wastewater capacity has not been verified.

This is not evidence the site lacks capacity. It means capacity is unknown.

That last distinction—**bad vs unknown**—is one of the strongest design principles you can adopt.

Separate facts, derived conclusions and value judgments

This directly follows Challenge 03's instruction to distinguish data from normative weights.
[filecite turn0file2](#)

Your interface should visibly distinguish:

Observed

Parcel area = 5,410 sq ft.
Zoning = X.
Flood overlay = no intersection in current GIS dataset.

Derived

Proposed unit count appears consistent with minimum-lot requirements.

Assumption

Construction cost = \$240/sq ft.

Unverified

Adequate sanitary capacity.

Normative

Transit access weight = 20%.

Conflicting sources

City and County official pages disagree on transfer-tax rate.

Requires professional review

Survey boundary / final zoning determination / structural conditions.

That kind of epistemic hygiene would be unusually strong for a hackathon.

The killer feature may be “What should I do Monday morning?”

Do not only produce analysis.

Produce the next action:

Next highest-value action: request Pittsburgh Water availability review.

Why: unresolved infrastructure capacity is the largest remaining feasibility uncertainty.

Inputs already available: parcel ID, proposed unit count.

Missing input: projected average and peak flow.

Who must prepare it: civil/MEP engineer.

Downstream decisions blocked by it: final site design, financial model, construction schedule.

Estimated cost of unresolved 60-day delay under current capital assumptions: \$131k.

Now you are moving from a dashboard to an operating tool.

Use historical permits as empirical priors

Challenge 02 asks for a regional observatory. Don't just map historic permits.

Use them to answer:

“For projects like mine, what actually happened?”

If sufficiently clean data are available:

- median days from application → first response;
- distribution of revision counts;
- time between review milestones;
- inspection failure/reinspection patterns;
- application abandonment;
- permit → occupancy duration;
- variance/hearing frequency;
- differences by project type;
- differences by review path;
- differences before/after process reforms.

Pittsburgh's OneStopPGH Insights already exposes current and historic permit/application/enforcement information, and the City has recently been consolidating datasets and workflows there. ⁵⁴

Therefore your opportunity is probably not:

“Let's scrape some permits onto a map.”

It is:

“Let's infer the project state machine and estimate the next transition.”

Three real projects would make a far stronger demo than citywide coverage

Pick three archetypes:

The easy parcel

A straightforward, by-right small residential infill project.

Your system should confidently say:

Few obvious barriers. Here's the fastest path.

The deceptive parcel

Looks attractive from zoning/price but triggers slope, stormwater, subdivision, utility, environmental or discretionary-review issues.

Your system should say:

This looks easy from the map, but here are the hidden dependencies.

The needed-but-not-financeable project

An affordable/adaptive-reuse project whose physical feasibility is strong but whose pro forma has a gap.

The system should say:

Regulation isn't the principal blocker. Capital structure is.

That final example is important. Otherwise your product will inherit the ideological assumption that every unbuilt unit is stuck because of zoning.

Sometimes the project simply does not earn enough revenue to cover land + construction + finance + operating costs.

What to measure during interviews

Build a spreadsheet—not a product—before the hackathon.

For every project discussed, record:

Field	Why
Project archetype	Enables comparisons
Units	Normalize friction
Total development cost	Cost denominator
Date site first investigated	Start of journey

Field	Why
Date site control obtained	Risk commitment
Date major approvals received	Entitlement duration
Date building permit issued	Permit duration
Date construction began	Mobilization
Date occupancy achieved	Total cycle
Number of agencies	Coordination load
Number of formal submissions	Process load
Number of revisions	Rework
Number of professional disciplines	Intelligence layers
Unplanned studies	Surprise
Biggest unknown discovered late	Product opportunity
Days waiting versus actively working	Queueing
Direct fees	Visible regulatory cost
Consultant expense	Hidden process cost
Carrying cost	Time cost
Major design changes	Rework cost
Root cause	Rule / data / coordination / physical / market
Could software have prevented it?	Automation hypothesis
Could faster processing cause harm?	Safety/equity check

Then you have something far more valuable than opinions.

You can say:

“Across 12 Pittsburgh projects totaling 184 units, we observed 2,430 cumulative project-days of preconstruction delay. Only 22% was active engineering; 38% was agency queueing; 17% was missing-information/rework; 13% was utility coordination; 10% other.”

I made those percentages up **as an example of the measurement you should produce**, not as findings. Do not pitch them until you have collected the evidence.

The hypotheses worth trying to kill

Go into Pittsburgh trying to disprove these rather than confirm them:

“Permitting is the bottleneck.”

Perhaps financing, utilities or construction costs dominate.

“The data exist but are fragmented.”

Perhaps some decisive data simply do not exist until someone does a study.

“Rules can be automated.”

Perhaps exceptions and discretionary interpretation dominate precisely the high-value projects.

“Developers don't know the requirements.”

Large developers may already know them perfectly; small builders may be the actual underserved user.

“Faster permitting means more housing.”

Perhaps approved projects still sit because the pro forma does not work.

“More housing means affordable housing.”

Production can alleviate scarcity while still producing units unaffordable to the lowest-income households.

“AI should make the decision.”

Perhaps users mainly need provenance, dependency management and earlier warnings.

“Government is the only fragmented actor.”

The private side—architects, engineers, title, lenders, contractors, insurers—may be equally fragmented.

If you kill three of your favorite assumptions before the hackathon, your eventual product will probably be substantially better.

The deepest opportunity

The sponsor briefs talk separately about permitting, housing data and typology. [filecite turn0file0](#)
[filecite turn0file1](#) [filecite turn0file2](#)

But I think there is one underlying problem:

Nobody possesses a continuously updated, source-grounded model of “what must be true for this parcel to become these homes.”

Today, that model is distributed across:

- a developer's head;
- a zoning lawyer's head;
- an architect's checklist;
- a civil engineer's drawings;
- utility correspondence;
- government databases;
- title documents;

- spreadsheets;
- PDFs;
- email threads;
- permit comments;
- hearing decisions;
- lender requirements;
- an inspector's observations.

The intelligence already exists.

It is just **fragmented across humans and institutions**.

That is probably the highest-value place for AI: not replacing the intelligence, but **making the distributed intelligence computationally visible**.

The house is ultimately made of concrete, wood, steel, drywall, glass and labor.

The *permission and confidence to build it* are made of information.

And right now, that information supply chain may be nearly as important a bottleneck as the physical one.

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